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Model Predictive Control with Linear Programming for the Strip Infeed in Hot Rolling Mills

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Abstract: A model-based predictive approach is proposed for the strip head motion control during the steel strip infeed in hot rolling finishing mills. The design is based on a nonlinear simplified mathematical model in a form of ordinary differential equations. It is already shown that this model captures the behavior of FEM models for the strip and the roll gap in an excellent manner. Then the simplified time variant linear model is considered and the discrete-time model is derived in a straightforward manner. In addition the property of flatness is exploited to derive an efficient formulation for the optimization problems. We develop an optimal controller based on a linear programming and compare it with standard quadratic programming. It is shown that the proposed linear program is both more efficient than the quadratic program and the performance is equivalent. Therefore, the linear approach is convenient for the real-time application even without high performance computers. The proposed controller is tested in a co-simulation environment using FEM element model for the strip integrated in HOTINT together with a highly nonlinear roll gap model. The simulation results show the excellent accordance of the controlled infeed process for both models (ODE and FEM), as well as the high performance of the closed loop.

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1. INTRODUCTION

In a hot rolling mill the incoming steel slabs are transformed into thin hot rolled coils. Functionally, the mill consists of two parts: roughing mill and finishing mill. The slabs are passed forwards and backwards through the roughing stand several times. During each pass the slab thickness is reduced by rolling it between two parallel rolls while moving the work rolls. Afterward the steel work piece passes through a series of finishing mill stands. One of the undesired effects, which can arise because of numerous uncontrollable factors, is strip camber (the strip forms a curve in the rolling plane) both in the roughing stand and in the finishing mill. The sources are for example: an asymmetric temperature distribution across the strip width (a temperature distribution of 100 K between both sides of the slab corresponds to a rolling force difference of about 22%, resulting in different thickness reductions on the two sides of the strip and thus leads to the strip camber Lukasson-Herzig (2007)), asymmetric input thickness profile, different wear etc. The curved strip will have not only a negative impact on the strip quality, but can even cause stops in the rolling process, due to *collision with side guides*. To avoid the collisions and therefore, quality reduction, different control strategies were proposed in recent decades. At the same time advanced control techniques, such as model predictive control, have been widely applied in industry. Model predictive control is one of few suitable methods for the process with different, sometimes contradictory, quality requirements and technical constraints.

1.1 State of the art of the camber/off-center modeling and control

The strip camber is a well-known problem in hot rolling. In most of the publications the strip camber is considered in the roughing mill and off-center, the distance between a center line of a strip and center line of the roll, is considered in finishing mill. The problem of the off-center control is often called steering control. These two effects are treated together or separately. In Lee and Choi (2014) and Kampmeijer et al. (2013) the camber is controlled at the roughing stand and steering control is implemented during rolling in the finishing mill. For camber/off-center analysis and control goals Nakajima's model is often used as a conventional model. In Okamura and Hoshino (1997) the parallel mill modulus control based on the Nakajima's model and usage of the bending force difference is proposed. In Kiyota et al. (2003) the conventional model is modified for the strip which is unconstrained both at the entry side and the exit side and the LQR control for the strip tail is proposed. Kiyota et al. (2003) assumes that only the pressure difference and the roll gap height can be measured and an observer is used to estimate the off-center value and its derivative. PID sliding mode

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control with an observer for the conventional model with tension constraint is proposed in Choi and Lee (2009). An observer-based guaranteed cost control with similar model is considered in Kim and Won (2013). Choi et al. (2008) shows the advantages of the model predictive control for the conventional linear model in hot rolling. In Schausberger et al. (2015) another mathematical model of the camber evolution of heavy plates is proposed and the optimization-based approach for reduction of the plate camber is considered in Schausberger et al. (2016). A simplified nonlinear model and model predictive control based on a quadratic program was presented in Galkina et al. (2016b) and Galkina et al. (2016a), Gafur et al. (2016) respectively.

1.2 Overview of the presented contribution

In this contribution an infeed process in finishing mill is considered. Taking into account the physical conditions we can divide the total rolling process in finishing mills into three stages: strip head has passed the upstream roll stand but does not reach yet the downstream roll stand (infeed process), see Fig. 1, strip under the tension between two roll stands and strip tail has passed the upstream roll stand, but does not reach yet the downstream roll stand. During the strip infeed the control goal is to keep the free strip head inside of the tolerance limits to avoid the rolling process interruption due to collision with the side guides (see Fig. 1) and to satisfy the strip quality requirements. In general, mathematical modeling of the strip infeed is a challenging task because the controlled system is in fact a distributed parameter system. Furthermore, measurements in this process are limited and most of the contributing factors are difficult to measure during the normal process at a plant. Thus, the control of this physical process is also challenging.

In the next Section the simplified nonlinear mathematical model in form of ordinary differential equations (ODE) is considered at first. This model is discretized with respect to time and then parameterized by the flat output to simplify the formulation of the optimization task. MPC (Model Predictive Control) based on a quadratic program was presented in Galkina et al. (2016b) and Gafur et al. (2016). MPC presented in Gafur et al. (2016) was successfully tested at the plant. This MPC has a relative short prediction horizon (0.6 s) and is implemented in a slow hardware environment. In the subsequent sections we present the formulation, which has much longer horizon (2.8 s), no waves generation and is applicable to the existing real-time system. In the third Section two model predictive strategies are proposed: stabilization of the optimal reference trajectory and step-wise recurrent generation of the new optimal trajectory. Both presented strategies are implemented as a quadratic program (QP) and linear program (LP) and then compared. It is shown that a model predictive control in form of LP is convenient for the real-time implementation in the strip infeed process, because it is more efficient in computation time and has the same dynamic performance. The last Section is devoted to the test of the designed MPC in the co-simulation environment including HOTINT with an integrated finite element model (FEM) of the strip and the roll gap model. The MPC performance with the FEM model is then compared to the simulation of MPC with ODE in MATLAB.

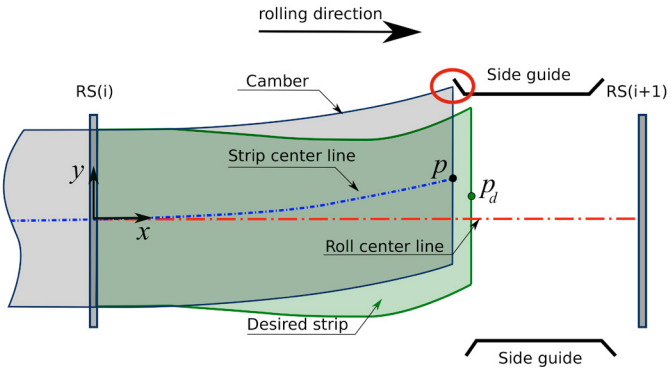


Fig. 1. Strip camber during infeed; RS(i) - upstream roll stand and RS(i+1) - downstream roll stand.

2. MATHEMATICAL MODEL

The mathematical model convenient for the describing the infeed process must be as simple as possible from one side and allows to describe the position of the tracked free strip head quite accurate. To satisfy these contradictory requirements some assumptions must be made, which simplify the modeling. Strip is assumed to be a rigid rod deformed in the rolling direction (the plastic deformation takes place in the roll gap only). The strip mass is neglected. The steel density depends mainly on the temperature, but not on the pressure. Therefore, the density is changed by cooling down during rolling process, but not by rolling forces in the roll gap. The strip entry and output width in roll stand is assumed to be constant. The entry strip wedge is linear. If an automatic roll gap control works sufficiently well, there is no output strip wedge. Also there is no velocity in the lateral direction. We also assume that the initial strip off-center, the difference between strip center line and roll center line in the roll gap (see Fig. 1), is constant and we do not consider the influence of the strip camber from the roughing stand. With these assumptions the strip camber will be generated due to a linear asymmetric output velocity profile and the strip moves in the rolling direction and rotates.

According to the assumptions from above a pure geometric modeling is suggested. We introduce a fixed coordinate system at the center of the roll gap (see Fig. 1). Motion of any point of the strip (for example, p in Fig. 1), leaving the roll gap with the velocity $[\,v\,(t)\;\;0\,]^{\mathrm{T}}$, with $v\,(t)$ - an output velocity of the strip in the roll gap, can be described by

$$\begin{bmatrix}\dot{x}(t)\\ \dot{y}(t)\end{bmatrix} = \begin{bmatrix}v(t)\\ 0\end{bmatrix} + \kappa(t)v(t)\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}\begin{bmatrix}x(t)\\ y(t)\end{bmatrix}, \quad (1)$$

where $\kappa(t)$ denotes a curvature at the origin of the chosen fixed coordinate system. The model (1) can be found in Gafur et al. (2016) and Galkina et al. (2016b), Galkina et al. (2016a). Equation (1) describes a motion of an arbitrary point of the strip before the strip enters downstream roll stand. In the presented model the curvature of the strip κ depends on numerous uncontrolled and immeasurable process factors as well as on the control action, i.e. leveling in the roll stand. Leveling u is proportional to the

position difference in the two hydraulic actuators pressing the rolls against the strip. Thereby, the strip deformation can be influenced. Although κ depends nonlinear on the leveling, the allowed leveling variation is small enough and the linearized relation can be considered. HOTINT with the FEM simulation also showed the linear dependency between curvature κ and leveling

$$\kappa = k_0 + k_1 u\,,$$

where k_0 and k_1 - parameters depending on an asymmetric strip thickness profile, strip thickness reduction, roll stand geometry etc. The presented ODE model (1) was tested with the FEM model integrated in HOTINT together with roll gap model and showed an excellent accuracy in the operating range. The model covers a sufficiently wide range of operating states and is nevertheless feasible for control purposes.

During the infeed control the following requirements must be satisfied:

(1) The strip head moves in a predefined target area before the next roll stand (avoiding the collision with the side guides).
(2) Straight entry of the strip in the downstream roll stand.
(3) The leveling must be kept in predefined limit.
(4) The rate of the wedge change must be kept in predefined limits (the leveling rate must be kept in predefined limit).

To satisfy these requirements a new input for the rate of leveling change is introduced and the simplified form of (1) for the subsequent MPC design is considered

$$\begin{aligned} \dot{x} &= \upsilon \\ \dot{y} &= (k_0 + k_1 u)\,\upsilon x \\ \dot{u} &= w\,. \end{aligned} \tag{2}$$

In (2) the state x is not influenceable anymore and the system is time-variant. State x and y are measured by a camera, which has the sampling time and the delay 20 ms. To take the delay into account, the simplest possibility is a discretization of (2) with the discretization time equal to the sampling time and including the delay into the discrete-time model.

2.1 Discrete time model

To simplify the calculation we set a new input

$$\hat{u} = k_1 u\,. \tag{3}$$

Using an exact discretization of (2) we get difference equations

$$\begin{aligned} x_{k+1} &= x_k + \upsilon\Delta t \\ y_{k+1} &= y_k + \frac{1}{6}\left(2\upsilon\Delta t + 3x_k\right)\hat{w}_k\upsilon\Delta t^2 + \frac{1}{2}\left(\hat{u}_k + k_0\right)\upsilon^2\Delta t^2 \\ &\quad + \left(\hat{u}_k + k_0\right)x_k\upsilon\Delta t \\ \hat{u}_{k+1} &= \hat{u}_k + \hat{w}_k\Delta t\,, \end{aligned} \tag{4}$$

where x_k and y_k describes the displacement of the middle point of the strip head, $\hat{w}_k = k_1 w_k$, Δt - is the sampling time, which is equal to the dead-time of the camera. Then, to simplify the formulation for the optimal trajectory generation and MPC, the discretized model will be parameterized by a flat output.

2.2 Parameterization by flat output

The idea of a discrete time flat system is the same as for time continuous systems - one is looking for a parameterization of all system variables in terms of a special set of independent variables called the flat outputs, such that the equations in (4) are satisfied (see, for example Sira-Ramirez and Agrawal (2004)). A suitable choice for the flat output h_k given by

$$h_{k+1} = \frac{(2\Delta t\upsilon + 3x_k)}{6}\upsilon\Delta t\hat{u}_{k+1} - y_{k+1}\,.$$

The system in the new coordinates reads as

$$\begin{aligned} x_{k+1} &= x_k + \upsilon\Delta t \\ h_{k+1} &= h_k - \frac{1}{2}\left(\upsilon\Delta t + (x_k + x_{k-1})\right)\hat{u}_k\upsilon\Delta t - \\ &\quad \frac{1}{2}k_0\left(\upsilon\Delta t + 2x_k\right)\upsilon\Delta t \\ \hat{u}_{k+1} &= \hat{u}_k + \hat{w}_k\Delta t\,. \end{aligned} \tag{5}$$

If the sequence h_k is specified, $\hat{u}_k$ can be calculated from the second equation in (5) and then w_k can be calculated from the third one. The system variables $[\,y_k\ \hat{u}_k\ \hat{w}_k\,]^{\mathrm{T}}$ can be defined from $[\,h_k\ h_{k+1}\ h_{k+2}\,]^{\mathrm{T}}$ whereby

$$\begin{bmatrix} y_k \\ \hat{u}_k \\ \hat{w}_k \end{bmatrix} = \mathbf{A}\left(x_{k-1},\, x_k,\, x_{k+1}\right)\begin{bmatrix} h_k \\ h_{k+1} \\ h_{k+2} \end{bmatrix} + \mathbf{b}\left(x_{k-1},\, x_k,\, x_{k+1}\right) \tag{6}$$

Details can be found in Galkina et al. (2016a).

3. MPC STRATEGIES

Although MPC is successfully applied to industrial problems since many years, it is mostly implemented in the process industries at the sufficiently 'slow' plants (see for example, Mayne et al. (2000)). Hot rolling in finishing mill is a relatively 'fast' process in comparison to petrochemical and related industries. The computational cost plays the crucial role for the MPC implementation in hot rolling finishing mills, where the optimization time depends on both, the chosen strategy for requirement fulfillment and the structure of the optimization program.

We compare two strategies for the model predictive control. The first one is a tracking problem, where the strip head follows a reference trajectory, was presented in Galkina et al. (2016a). The optimal reference trajectory for the infeed is generated only once, when the measurement data are available. Then the model predictive controller stabilizes the desired trajectory. The second strategy is new and is faster than the first one. At the same time it will be shown that implementation of LP for both strategies has an advantage in computational cost comparing to QP, which is often implemented as a standard formulation (see Choi et al. (2008), Rao and Rawlings (2000)). Another MPC based on a QP and model (1) for the infeed process in the finishing mill, which was successfully tested at the plant, can be found in Gafur et al. (2016).

3.1 Optimal reference trajectory and MPC for stabilization of the reference trajectory (MPC SRT)

In this part the optimal reference trajectory generation and model predictive control for the stabilization of this

trajectory is considered. At first we formulate the trajectory generation and MPC in the form of QP and compare this formulation with LP. The parameters k_0 and k_1 are assumed to be known. In comparison to formulation in Galkina et al. (2016a) we restrict also a sign of w_k depending on sign of k_0. This is made to avoid a generation of the height waves along the strip length which happens in both existing solutions (Galkina et al. (2016a), Gafur et al. (2016)) and reduces the strip quality.

The optimal reference trajectory generation is based on the minimization of $|w_k|$ without sign changes (see (7)). Under the given process requirements, it can be formulated as a quadratic optimization program (not in a standard form)

$$\begin{array}{rrcccl} & & \min\limits_{\mathbf{h}} \epsilon^2 & & & \\ & 0 & \leq \epsilon \leq & W_{max} & & \\ & -U_{max} & \leq u_k \leq & U_{max}\,, & k=2,..,N_0-1 & \\ & -Y_{max} & \leq y_k \leq & Y_{max}\,, & k=3,..,N_0 & \\ \text{if } k_0>0: & -\epsilon & \leq w_k \leq & 0\,, & k=1,..,N_0-2 & \\ \text{if } k_0<0: & 0 & \leq w_k \leq & \epsilon\,, & k=1,..,N_0-2 & \\ & y_N & = & y_{N-1} & & \end{array} \tag{7}$$

with the optimization vector $\mathbf{h} = [\,h_3\ ...\ h_N\ \epsilon\,]^T$, the system states $[\,y_k\ u_k\ w_k\,]^{\mathrm{T}}$, which are calculated from (6) and (3) and the number N_0 of steps from camera to the downstream roll stand. h_1 and h_2 depend on the initial state of (5). The number of steps N_0 can be calculated from the geometry of the finishing mill. It is assumed that the problem is feasible. The equivalent linear program with l_1 cost function is

$$\begin{array}{rrcccl} & & \min\limits_{\mathbf{h}} \epsilon & & & \\ & 0 & \leq \epsilon \leq & W_{max} & & \\ & -U_{max} & \leq u_k \leq & U_{max}\,, & k=2,..,N_0-1 & \\ & -Y_{max} & \leq y_k \leq & Y_{max}\,, & k=3,..,N_0 & \\ \text{if } k_0>0: & -\epsilon & \leq w_k \leq & 0\,, & k=1,..,N_0-2 & \\ \text{if } k_0<0: & 0 & \leq w_k \leq & \epsilon\,, & k=1,..,N_0-2 & \\ & y_N & = & y_{N-1} & & \end{array} \tag{8}$$

To provide robustness against parameter inexactness a model predictive control is proposed. MPC solves the optimization task for minimization of the error $e_y = y_k - y_r$, where y_r denotes an optimal trajectory, which was generated at the most first step. The minimization of the error e_y and w_k under the given process requirements can be formulated as a quadratic optimization problem

$$\begin{array}{rcccl} & & \min\limits_{\mathbf{h}} \epsilon_1^2+\epsilon_2^2 & & \\ 0 & \leq \epsilon_1 \leq & W_{max} & & \\ 0 & \leq \epsilon_2 \leq & E_{y,max} & & \\ -U_{max} & \leq u_k \leq & U_{max}\,, & k=2,..,N_0-1-n & \\ -Y_{max} & \leq y_k \leq & Y_{max}\,, & k=3,..,N_0-n & \\ -\epsilon_1 & \leq w_k \leq & \epsilon_1\,, & k=1,..,N_0-2-n & \\ -\epsilon_2 & \leq y_k - y_{r,n+1} \leq & \epsilon_2\,, & k=3,..,N_0-n & \end{array} \tag{9}$$

with optimization vector $\mathbf{h} = [\,h_3\ ...\ h_N\ \epsilon_1\ \epsilon_2\,]^{\mathrm{T}}$. The actual prediction horizon, $N = N_0 - n$, is decreasing with each new step n. The equivalent linear program is

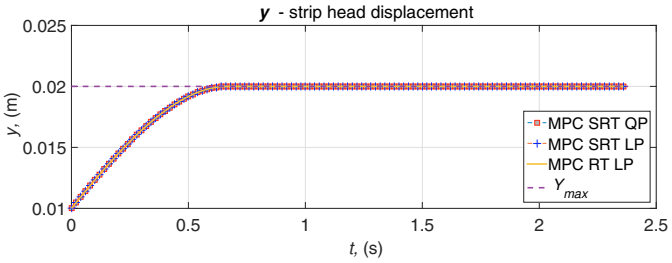


Fig. 2. Strip head position during infeed

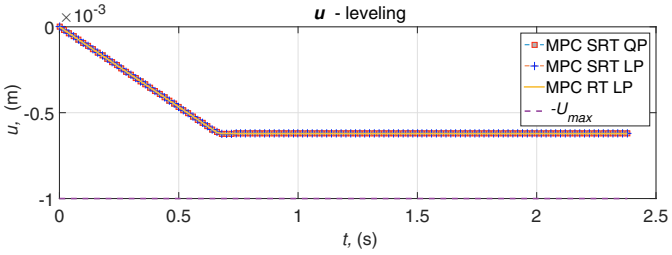


Fig. 3. Leveling during infeed.

$$\begin{array}{rcccl} & & \min\limits_{h} \epsilon_1+\epsilon_2 & & \\ 0 & \leq \epsilon_1 \leq & W_{max} & & \\ 0 & \leq \epsilon_2 \leq & E_{y,max} & & \\ -U_{max} & \leq u_k \leq & U_{max}\,, & k=2,..,N_0-1-n & \\ -Y_{max} & \leq y_k \leq & Y_{max}\,, & k=3,..,N_0-n & \\ -\epsilon_1 & \leq w_k \leq & \epsilon_1\,, & k=1,..,N_0-2-n & \\ -\epsilon_2 & \leq y_k - y_{r,n+1} \leq & \epsilon_2\,, & k=3,..,N_0-n\,. & \end{array} \tag{10}$$

Simulations with the nonlinear ODE model in MATLAB, both MPC SRT QP and MPC SRT LP, show the same dynamic performance, see Fig. 2, Fig. 3 and 4.

3.2 MPC with stepwise recurrent generation of the optimal trajectory (MPC RT)

The strategy of the stepwise recurrent generation of the optimal trajectory for the new available measurement data is based on the recurrent solving of (7) (substituting N instead of N_0) or (8). This strategy is not only simpler in formulation, because there is no need to solve the task (9) or (10), but it is faster in comparison to solving the optimization program for the stabilization of the reference trajectory. In Fig. 2 it is shown that there is no difference in the strip head middle point displacement for two different strategies, when the strip head motion is simulated in MATLAB based on the nonlinear ODE model (1) and the model parameters k_0 and k_1 are known exactly. The QP for the stabilization of the reference trajectory (9) is solved by MATLAB Optimization Toolbox solver "quadprog" with active-set method. The stabilization with linear formulation (10) is solved by "linprog" with simplex method and by free linear programming solver for MATLAB based on the revised simplex method "lp_solve" . The MPC with recurrent trajectory generation in linear form (8) is solved with "lp_solve". Fig. 3 and Fig. 4 show the corresponding leveling and rate of the leveling change. In Fig. 5 the optimization time for both MPC strategies and both formulations (QP and LP) are shown. It can be seen that the recurrent trajectory combined with a linear program formulation and appropriate choice of the solver

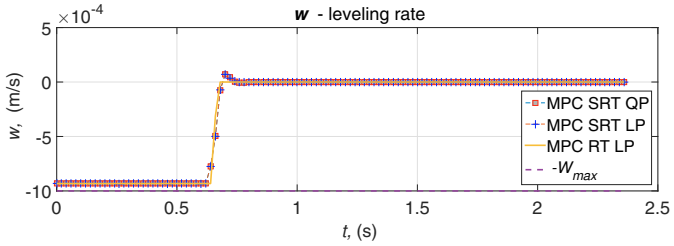


Fig. 4. Rate of the leveling change during infeed.

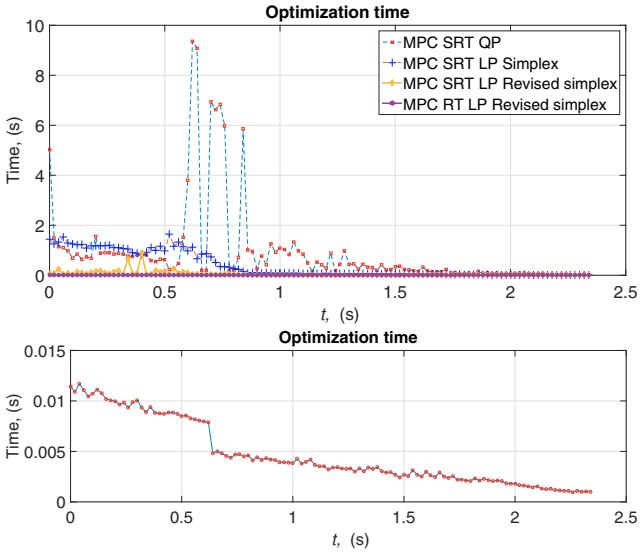


Fig. 5. Optimization time: comparison (up), MPC RT LP with revised simplex method (down)

demonstrates the lowest computational cost. The revised simplex method is a scheme for ordering the computations required for the simplex method so that unnecessary calculations are avoided. Therefore, this method is a better choice for solving the presented problem (see, for example, Luenberger and Ye (2008)). The execution time of the linear program, which is solved with revised simplex method, is smaller than the camera sampling time (see Fig. 5). Of course, the optimization time depends on the prediction horizon N. "lp_solve" execution time is appropriate for $N \leq 140$. To cover longer distance another solver realization must be implemented, for example as a C++ program. Another important advantage of the recurrent generation of the optimal trajectory is the robustness with respect to variation of k_0. The choice of $E_{y,max}$ strongly depends on k_0, measuring device resolution and measurement disturbance. There exist extensions for the linear and quadratic MPC approach, where the parameters k_0 and k_1 are estimated online, but these extensions are not part of this contribution.

4. CO-SIMULATION WITH FINITE ELEMENT MODEL OF THE STRIP

The performance of the controllers developed above is tested by help of co-simulation of complex FEM models (see Vetyukov et al. (2016a), Vetyukov et al. (2016b)) and Simulink Matlab. The present version of the simulation software is capable of modeling controlled motion of the rolled metal strip between pairs of subsequent roll stands.

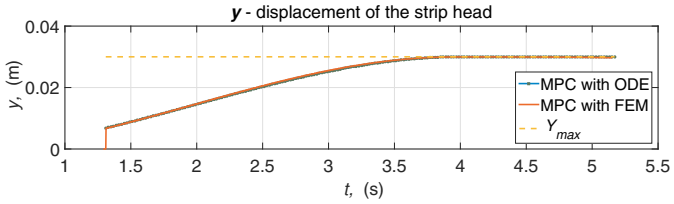


Fig. 6. Displacement of the strip head with MPC based on linear program

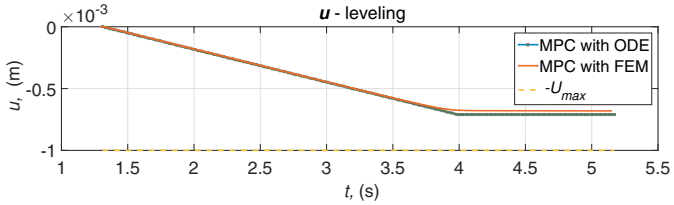


Fig. 7. Leveling with MPC based on linear program

4.1 Co-simulation environment

The software includes a finite element strip model and a roll gap model. The model of the strip is implemented in Hotint (see Gerstmayr (2009)), which is also responsible for providing access to the general model settings for all parts of the simulation software. The control system is realized in Simulink Matlab. The data exchange between Hotint and Simulink is realized through TCP/IP.

4.2 Co-simulation results

The following parameters were set for the co-simulation:

- Young's modulus E of the steel at about 1000°C, $6.6 + e10$ [Pa] (Steel ST37)
- Strip width, 1 [m]
- Input thickness profile values (at the most left and right strip sides), [0.0351 0.0349] [m]
- Output thickness profile values (at the most left and right strip sides), [~ 0.0235 ~ 0.0235] [m]
- Slab entry velocity profile values (at the most left and right strip sides), [0.778305 0.778305] [m/s]
- Distance from the upstream roll stand to the measurable area, 1.5 [m]

In the co-simulation environment the strip head displacement is measured in the lateral and rolling direction by virtual sensors and the parameters k_0 and k_1 are known. In Fig. 6, 7 and 8 the system states with FEM co-simulation environment and ODE simulation are compared. The performance of MPC with ODE, as well as MPC with FEM model of the strip match very well.

In a real plant the direct measurement of the strip head is not available. The parameters k_0 and k_1 are unknown and the water steam disturbs the measurement significantly. The extensions, where the strip head position and the unknown parameters k_0, k_1 are estimated from the disturbed camera measurements position are not part of this contribution. The MPC based on QP and estimated parameters (see Gafur et al. (2016)) was successfully tested at the plant.

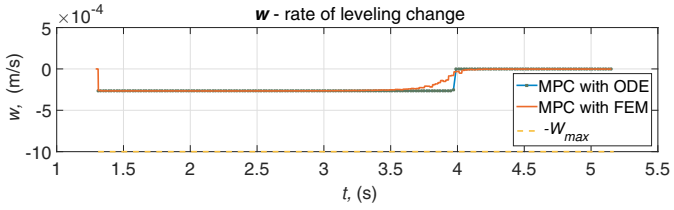


Fig. 8. Rate of leveling change with MPC based on linear program

CONCLUSION

In this contribution the model-based predictive control applying linear programming techniques is proposed for the strip head motion control during the infeed process in hot rolling finishing mill. For the MPC design the nonlinear simplified mathematical model describing the strip behavior during the infeed is considered. The simplified model is discretized and then parameterized with the flat output to simplify the MPC formulation. It is shown that MPC based on the recurrent generation of the optimal trajectory in the form of a linear program can be convenient for the real-time application in the hot rolling process for the infeed control. The proposed MPC was tested in the free available software package HOTINT with two integrated commercial models of the metal strip and the roll gap. The MPC with the simplified mathematical model and the MPC with the complex FEM model of the strip show a very good match.

In the future work it is still necessary to reduce the computational cost to guarantee that the designed MPC is convenient for the wide range of finishing mill configurations even without high performance computers.

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REFERENCES

Choi, I., Rossiter, J., Chung, J., and Fleming, P. (2008). An MPC strategy for hot rolling mills and applications to strip threading control problems. In *Proceedings of the 17th IFAC World Congress*, 1661–1662.

Choi, Y.J. and Lee, M. (2009). PID Sliding mode control for steering of lateral moving strip in hot strip rolling. *International Jounal of Control, Automation and Systems*, 7, 399–407.

Gafur, I., Galkina, A., Schlacher, K., and Lorenz, K.A. (2016). *Modelpraediktive Regelung des Bandkopfes in der Warmbandstrasse.* Patent. 2015E21438 DE. *To appear.*

Galkina, A., Gafur, I., and Schlacher, K. (2016a). Modellpraediktive Regelung fuer das Bandeinfaedeln in der Warmwalzstrasse. *BHM Berg- und Huettenmaennische Monatshefte*, 161(11), 520–525. doi:10.1007/s00501-016-0539-6.

Galkina, A., Gafur, I., and Schlacher, K. (2016b). Nonlinear model of the two dimensional plane strip in the hot rolling process. In *17th IFAC Symposium on Control, Optimization and Automation in Mining, Mineral and Metal Processing*, 41. Vienna, Austria.

Gerstmayr, J. (2009). Hotint a c++ environment for the simulation of multibody dynamics systems and finite elements. In *Proceedings of the Multibody Dynamics 2009 ECCOMAS Thematic Conference*, 1– 20.

Kampmeijer, L., Hol, C., de Roo, J., and Spelbosi, E. (2013). Strip tracking measurement and control in hot strip rolling. In *9th International Conference of Rolling, Metallurgia Italiana*, 29–34.

Kim, K. and Won, S. (2013). Observer-based guaranteed cost control for lateral motion of a strip in hot rolling process. In *10th IEEE International Conference on Control and Automation (ICCA)*, 1479 –1484.

Kiyota, T., Matsumoto, H., Adachi, Y., Kondo, E., Tsuji, Y., and Aso, S. (2003). Tail crash control in hot strip mill by LQR. In *Proceedings of the American Control Conference, Denver*, 3049–3054.

Lee, J. and Choi, Y.J. (2014). Development of camber and steering control system in hot strip mill. In *2014 IEEE Emerging Technology and Factory Automation (ETFA)*, p. 1.

Luenberger, D. and Ye, Y. (2008). *Linear and Nonlinear Programming.* Springer, 233 Spring Street, New York, NY 10013, USA, 4 edition.

Lukasson-Herzig, A. (2007). *Optimierung der Stahlbandgeometrie im Hinblick auf den Bandsaebel in Warmbreitbandstrassen.* Ph.D. thesis, RWTH Aachen University.

Mayne, D., Rawlings, J., Rao, C., and Scokaert, P. (2000). Constrained model predictive control: Stability and optimality. *Automatica*, 36, 789–814.

Okamura, Y. and Hoshino, I. (1997). State feedback control of the strip steering for aluminum hot rolling mills. *Control Engineering Practice*, 5(8), 1035–1042.

Rao, C. and Rawlings, J. (2000). Linear programming and model predictive control. *Journal of Process Control*, 10, 283–289.

Schausberger, F., Steinboeck, A., and Kugi, A. (2015). Mathematical modeling of the contour evolution of heavy plates in hot rolling. *Applied Mathematical Modelling*, 39(15), 4534–4547. doi:10.1016/j.apm.2015.01.01.

Schausberger, F., Steinboeck, A., and Kugi, A. (2016). Optimization-based reduction of contour errors of heavy plates in hot rolling. *Journal of Process Control*, 47, 150–160.

Sira-Ramirez, H. and Agrawal, S. (2004). *Differentially flat systems.* Marcel Dekker, New York.

Vetyukov, Y., Gruber, P., and Krommer, M. (2016a). Nonlinear model of an axially moving plate in a mixed Eulerian-Largangian framework. *Acta Mechanica*, 27(10), 2831–2842.

Vetyukov, Y., Gruber, P., Krommer, M., Gerstmayr, J., Gafur, I., and Winter, G. (2016b). Mixed eulerian-lagrangian description in materials processing: deformation of a metal sheet in a rolling mill. *International Journal for Numerical Methods in Engineering*, 1–20. doi:10.1002/nme.5314. Online.